Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem

- Sum Rule, Product Rule, and Bayes' Theorem
  Reading • 15 min
- Sum Rule
  Reading • 4 min
- Product Rule (Chain Rule)
  Reading • 2 min
- Bayes' Theorem
  Reading • 43 min

Lesson 2: Gaussian Distribution

- Univariate Gaussian Distribution
  Reading • 4 min
- Gaussian Distribution: Foundations and Applications
  Reading • 15 min
- Multivariate Gaussian Distribution
  Reading • 10 min
- **Conditional and Marginal Multivariate Gaussian Distributions**
  Reading • 45 min
- Product of Gaussian Densities
  Reading • 26 min

Lesson 3: Inference Techniques

- Bayesian Inference
  Reading • 1h 10m
- Latent-Variable Models
  Reading • 50 min
- Check Your Knowledge: Inference Techniques
  Graded Assignment • 10 min

Module 4
Introduction to Model Evaluation and Optimization

Conditional and Marginal Multivariate Gaussian Distributions

0:05 / 2:53 2x

Let's break down the concepts of marginalization and conditioning for multivariate Gaussian distributions, using the joint Gaussian distribution for two multivariate random variables X and Y.

Explore the fundamentals of multivariate Gaussian distributions and conditional Gaussian distributions with *Conditional Multivariate Gaussian, In Depth*. This resource covers key concepts like the mean vector, covariance matrix, and conditional distributions, with clear explanations and examples to deepen your understanding of these essential data science concepts.

Joint Gaussian Distribution

Given two multivariate random variables X and Y, the joint distribution is Gaussian and can be written as:

$$p(x, y) = \mathcal{N}\left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}\right)$$

Here:

- μx and μy are the mean vectors for X and Y respectively.
- Σxx and Σyy are the covariance matrices of X and Y.
- $\Sigma xy = Cov[X, Y]$ and $\Sigma yx = Cov[Y, X]$ represent the cross-covariance matrices between X and Y.

Conditional Gaussian Distributions

The conditional distribution of one set of variables given another in a multivariate Gaussian distribution is also Gaussian. Knowing some variables (conditioning on them) leads to a Gaussian distribution for the remaining variables, with a mean that shifts according to the known values and a reduced covariance.

The conditional distribution p(X|Y=y) is Gaussian and is given by:

$$p(X \mid Y = y) = \mathcal{N}\left(\mu_{x|y}, \Sigma_{x|y}\right)$$

Where:

- The conditional mean is:

$$\Sigma_{x|y} = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$$

This represents the reduced uncertainty in X after observing Y.

$$p(X) = \int p(X, Y)dy = \mathcal{N}\left(X \mid \mu_x, \Sigma_{xx}\right)$$

Similarly, the marginal distribution of Y can be obtained by integrating out X:

$$p(Y) = \int p(X, Y)dx = \mathcal{N}\left(Y \mid \mu_y, \Sigma_{yy}\right)$$

If you'd like to reinforce the concepts covered in the lesson, this optional video, Marginal & Conditional for the Multivariate Normal | Full Derivation, provides a visual explanation of Support Vector Machines (SVMs). It complements the written material by illustrating key ideas like margins, decision boundaries, and kernel tricks through clear examples and step-by-step guidance.

Applying Support Vector Machines to Real-World Data

For a practical application of the concepts covered in this lesson, refer to Example 6.6 on page 200 in *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong. This example demonstrates how Support Vector Machines (SVMs) are used to classify data, highlighting key concepts such as decision boundaries, margins, and kernel functions.

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